HW #5 Problem 1(b): Continuing From Your Amplitudes

Picking Up After the Feynman Rules

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1 Where You Are

Your Setup

Process: $q(p_1, c_i) + g(p_2, a) \to q(p_4, c_4^\dagger) + g(p_3, a^{b*})$

You wrote out M_t with all delta functions and internal momentum integration:

$$-iM_t = \int \frac{d^4q}{(2\pi)^4}\, \bar{u}(4)\, c_4^\dagger \left(\frac{-ig_s}{2}\lambda^a\gamma^\mu\right) \varepsilon_\mu(2)\, a\, \frac{i(\not{q}+m)}{q^2-m^2}\, \varepsilon_\nu^*(3)\, a^{b*} \left(\frac{-ig_s}{2}\lambda^b\gamma^\nu\right) u(1)\, c_i \times (2\pi)^4\delta^4(p_1 - q - p_3)\,(2\pi)^4\delta^4(q + p_2 - p_4) \tag{1}$$

You noted: masses drop ($m \to 0$), interference terms cancel. Now we continue.

2 Step 1: Integrate Over q (Collapse the Delta Functions)

The first delta function $\delta^4(p_1 - q - p_3)$ fixes:

$$\boxed{q = p_1 - p_3} \tag{2}$$

Check with the second delta: $\delta^4(q + p_2 - p_4) = \delta^4(p_1 - p_3 + p_2 - p_4)$. This is just overall momentum conservation $p_1 + p_2 = p_3 + p_4$. ✓

The $d^4q/(2\pi)^4$ integration eats one $(2\pi)^4\delta^4$, leaving the overall momentum-conservation delta (which we erase as per Griffiths' rules to get $-i\mathcal{M}$).

With $m = 0$:

$$q^2 = (p_1 - p_3)^2 = \hat{t} \tag{3}$$

So the t-channel amplitude simplifies to:

$$\boxed{M_t = \frac{-g_s^2}{4\hat{t}}\, \bar{u}(4)\, c_4^\dagger\, (\lambda^a\gamma^\mu)\, \varepsilon_\mu(2)\, a\, \not{q}\, \varepsilon_\nu^*(3)\, a^{b*} \left(\lambda^b\gamma^\nu\right) u(1)\, c_i} \tag{4}$$

where $q = p_1 - p_3$ and I've collected the factors: two vertices give $(-ig_s/2)^2 = -g_s^2/4$, the propagator numerator gives $i\not{q}$ (with $m = 0$), and the propagator denominator gives $1/\hat{t}$.

By the same procedure for the s-channel, with $q = p_1 + p_2$, $q^2 = \hat{s}$:

$$\boxed{M_s = \frac{-g_s^2}{4\hat{s}}\, \bar{u}(4)\, c_4^\dagger \left(\lambda^b\gamma^\nu\right) \varepsilon_\nu^*(3)\, a^{b*}\, \not{q}\, \varepsilon_\mu(2)\, a\, (\lambda^a\gamma^\mu)\, u(1)\, c_i} \tag{5}$$

(Note: the λ matrices swap order compared to M_t because the quark hits the gluon vertices in opposite order.)

3 Step 2: Factor Color From Dirac

Why Factorization Works

The λ^a matrices are 3×3 (act on color indices $i, j = 1, 2, 3$). The γ^μ matrices are 4×4 (act on spinor indices). They live in **different vector spaces**, so they commute with each other. We can pull all the color pieces out front.

***t*-channel:** Reading along the color indices from c_i to $c_4^\dagger$:

At the bottom vertex: quark color $i \to m$ via λ^b_{mi}

At the top vertex: quark color $m \to j$ via λ^a_{jm}

Chain: $\sum_m \lambda^a_{jm}\lambda^b_{mi} = (\lambda^a\lambda^b)_{ji}$

Gluon colors: a^{b*} at p_3 vertex, a at p_2 vertex (these contract with the λ color indices and go into the color factor upon summing).

So:

$$M_t = \frac{-g_s^2}{4}\,(\lambda^a\lambda^b)_{ji}\,(a^{b*})(a) \;\times\; \underbrace{\frac{1}{\hat{t}}\,\bar{u}(4)\gamma^\mu\varepsilon_\mu(2)\,\not{q}\,\gamma^\nu\varepsilon^*_\nu(3)\,u(1)}_{D_t} \tag{6}$$

***s*-channel:** At left vertex: $i \to m$ via λ^a_{mi}. At right vertex: $m \to j$ via λ^b_{jm}. Chain: $(\lambda^b\lambda^a)_{ji}$.

$$M_s = \frac{-g_s^2}{4}\,(\lambda^b\lambda^a)_{ji}\,(a^{b*})(a) \;\times\; \underbrace{\frac{1}{\hat{s}}\,\bar{u}(4)\gamma^\nu\varepsilon^*_\nu(3)\,\not{q}\,\gamma^\mu\varepsilon_\mu(2)\,u(1)}_{D_s} \tag{7}$$

Factorized Amplitudes

$$M_t = \frac{-g_s^2}{4}\,(\lambda^a\lambda^b)_{ji}\,(a^{b*}\cdot a) \;\times\; D_t \tag{8}$$

$$M_s = \frac{-g_s^2}{4}\,(\lambda^b\lambda^a)_{ji}\,(a^{b*}\cdot a) \;\times\; D_s \tag{9}$$

where D_t and D_s are **identical** to the QED Compton Dirac structures (with $e^2 \to 1$).
The gluon external color factors $(a^{b*}\cdot a)$ will be summed over when we do the color averaging — they just pick out specific gluon states.

4 Step 3: Square and Use Your QED Result

As you wrote:

$$|M|^2_{\text{QCD}} = |M_s|^2 + |M_t|^2 + \cancel{M_s M_t^* + M_s^* M_t} \tag{10}$$

The interference drops because the **spin/polarization** interference vanishes at $m = 0$ (as you showed in QED). Good.

Each squared term factorizes as:

$$|M_t|^2 = \frac{g_s^4}{16} \, |(\lambda^a \lambda^b)_{ji}|^2 \;\times\; |D_t|^2 \tag{11}$$

$$|M_s|^2 = \frac{g_s^4}{16} \, |(\lambda^b \lambda^a)_{ji}|^2 \;\times\; |D_s|^2 \tag{12}$$

Now spin/polarization average (factor $\frac{1}{4}$: 2 quark spins $\times$ 2 gluon polarizations):

$$\frac{1}{4} \sum_{\text{spins,pols}} |D_t|^2 = -\frac{2\hat{s}}{\hat{t}}, \qquad \frac{1}{4} \sum_{\text{spins,pols}} |D_s|^2 = -\frac{2\hat{t}}{\hat{s}} \tag{13}$$

These are exactly your QED results (the $-2g_e^4(\hat{s}/\hat{t}+\hat{t}/\hat{s})$ you computed, split into the two channels, with g_e^4 stripped off).

5 Step 4: Color Factor

Now we sum/average over colors. This is the new piece beyond QED.

What We Average/Sum Over

- Average over incoming quark color i: $\frac{1}{N_c} = \frac{1}{3}$
- Average over incoming gluon color a: $\frac{1}{N_c^2-1} = \frac{1}{8}$
- Sum over outgoing quark color $j = 1, 2, 3$
- Sum over outgoing gluon color $b = 1, \ldots, 8$

Total averaging factor: $\frac{1}{3} \times \frac{1}{8} = \frac{1}{24}$

The color factor for the $|t\text{-channel}|^2$ is:

$$\mathcal{C} = \frac{1}{24} \sum_{i,j=1}^{3} \sum_{a,b=1}^{8} \left| (\lambda^a \lambda^b)_{ji} \right|^2 \tag{14}$$

Step 4a. Use $\lambda^{a\dagger} = \lambda^a$ (Hermitian):

$$|(\lambda^a \lambda^b)_{ji}|^2 = (\lambda^a \lambda^b)_{ji} \cdot [(\lambda^a \lambda^b)_{ji}]^* = (\lambda^a \lambda^b)_{ji} \cdot (\lambda^b \lambda^a)_{ij} \tag{15}$$

Step 4b. Sum over $i, j \to$ trace:

$$\sum_{i,j} (\lambda^a \lambda^b)_{ji} (\lambda^b \lambda^a)_{ij} = \mathrm{Tr}\Big[\lambda^a \lambda^b \lambda^b \lambda^a\Big] \tag{16}$$

So:

$$\mathcal{C} = \frac{1}{24} \sum_{a,b} \mathrm{Tr}\left[\lambda^a \underbrace{\lambda^b \lambda^b}_{\text{sum over } b \text{ first}} \lambda^a\right] \tag{17}$$

Step 4c. Evaluate $\sum_b \lambda^b \lambda^b$:

This is the quadratic Casimir $\sum_b T^b T^b = C_F \mathbb{I}$ with $T^b = \lambda^b/2$:

$$\sum_{b=1}^{8} \lambda^b \lambda^b = 4 C_F \, \mathbb{I}_3 = 4 \times \frac{4}{3} \times \mathbb{I}_3 = \frac{16}{3} \, \mathbb{I}_3 \tag{18}$$

where $C_F = \frac{N_c^2 - 1}{2N_c} = \frac{8}{6} = \frac{4}{3}$.

Explicit Verification

Check the $(1,1)$ element: $\sum_{b=1}^{8} \sum_{k=1}^{3} |\lambda^b_{1k}|^2$

λ^b	nonzero entries in row 1	$\lvert\lambda^b_{1k}\rvert^2$
λ^1	$\lambda^1_{12} = 1$	1
λ^2	$\lambda^2_{12} = -i$	1
λ^3	$\lambda^3_{11} = 1$	1
λ^4	$\lambda^4_{13} = 1$	1
λ^5	$\lambda^5_{13} = -i$	1
λ^6	—	0
λ^7	—	0
λ^8	$\lambda^8_{11} = 1/\sqrt{3}$	1/3
	Total:	16/3 ✓

Step 4d. Substitute back:

$$\mathcal{C} = \frac{1}{24} \sum_a \mathrm{Tr}\left[\lambda^a \cdot \frac{16}{3} \, \mathbb{I} \cdot \lambda^a\right] = \frac{1}{24} \cdot \frac{16}{3} \sum_a \mathrm{Tr}(\lambda^a \lambda^a) \tag{19}$$

Step 4e. Use $\mathrm{Tr}(\lambda^a \lambda^a) = 2\delta^{aa} = 2$ (no sum), so:

$$\sum_{a=1}^{8} \mathrm{Tr}(\lambda^a \lambda^a) = 2 \times 8 = 16 \tag{20}$$

Step 4f. Final:

$$\mathcal{C} = \frac{1}{24} \times \frac{16}{3} \times 16 = \frac{256}{72} = \frac{32}{9} \tag{21}$$

Color Factor

$$\boxed{\mathcal{C}_t = \mathcal{C}_s = \frac{32}{9}} \tag{22}$$

The s-channel gives the same result by $a \leftrightarrow b$ symmetry.

6 Step 5: Put It All Together

Combining everything:

$$\overline{|\mathcal{M}|^2}_{\text{QCD}} = \underbrace{\frac{g_s^4}{16}}_{\text{coupling}} \times \underbrace{\frac{32}{9}}_{\text{color}} \times \underbrace{\left[-2\left(\frac{\hat{s}}{\hat{t}} + \frac{\hat{t}}{\hat{s}}\right)\right]}_{\text{spin (your QED result with } g_e^4=1)} \tag{23}$$

Simplify the coupling × color prefactor:

$$\frac{g_s^4}{16} \times \frac{32}{9} = \frac{32\, g_s^4}{144} = \frac{2g_s^4}{9} \tag{24}$$

So:

$$\overline{|\mathcal{M}|^2}_{\text{QCD}} = \frac{2g_s^4}{9} \times \left[-2\left(\frac{\hat{s}}{\hat{t}} + \frac{\hat{t}}{\hat{s}}\right)\right] = \frac{-4g_s^4}{9}\left(\frac{\hat{s}}{\hat{t}} + \frac{\hat{t}}{\hat{s}}\right) = \frac{-4g_s^4(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}\hat{t}} \tag{25}$$

Cross-Check Against Your QED Result

Your QED result: $\overline{|\mathcal{M}|^2}_{\text{QED}} = -2g_e^4(\hat{s}^2+\hat{t}^2)/(\hat{s}\hat{t})$.
Ratio:

$$\frac{\overline{|\mathcal{M}|^2}_{\text{QCD}}}{\overline{|\mathcal{M}|^2}_{\text{QED}}} = \frac{-4g_s^4/9}{-2g_e^4} = \frac{2g_s^4}{9g_e^4} = \frac{2(4\pi\alpha_s)^2}{9(4\pi\alpha)^2} = \frac{2\alpha_s^2}{9\alpha^2} \tag{26}$$

Coupling replacement $\alpha^2 \to \alpha_s^2$ times a pure color factor 2/9. Makes sense.

7 Step 6: Cross Section

Using $g_s^2 = 4\pi\alpha_s$, so $g_s^4 = 16\pi^2\alpha_s^2$:

$$\overline{|\mathcal{M}|^2}_{\text{QCD}} = \frac{-4 \times 16\pi^2\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}\hat{t}} = \frac{-64\pi^2\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}\hat{t}} \tag{27}$$

Cross section formula (massless $2 \to 2$ scattering):

$$\frac{d\hat{\sigma}}{d\hat{t}} = \frac{\overline{|\mathcal{M}|^2}}{16\pi\hat{s}^2} \tag{28}$$

$$\begin{aligned}\frac{d\hat{\sigma}}{d\hat{t}} &= \frac{-64\pi^2\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9 \times 16\pi \times \hat{s}^3\hat{t}} \\ &= \frac{-4\pi\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}^3\hat{t}}\end{aligned} \tag{29}$$

Final Answer

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{t}}(qg \to qg)\right|_{2\text{ diag}} = \frac{-4\pi\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}^3\hat{t}}} \tag{30}$$

This is positive because $\hat{t} < 0$ for physical scattering.

Equivalently:

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{QCD}} = \frac{2\alpha_s^2}{9\alpha^2} \times \left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{your QED}}} \tag{31}$$

8 Summary: What Each Piece Came From

Step-by-Step Accounting

Piece	Factor	Origin
QCD vertices	$g_s^4/16$	Two vertices, each $g_s/2$
Dirac traces	$-2(\hat{s}/\hat{t} + \hat{t}/\hat{s})$	Identical to your QED calc
Avg initial quark	$1/3$	3 colors equally likely
Avg initial gluon	$1/8$	8 gluon states
$\sum_{i,j,a,b} \lvert(\lambda^a\lambda^b)_{ji}\rvert^2$	$256/3$	$\mathrm{Tr}[\lambda^a\lambda^b\lambda^b\lambda^a]$
Color factor	$32/9$	$(1/24) \times (256/3)$
Coupling $\times$ color	$2g_s^4/9$	$(g_s^4/16) \times (32/9)$
$\overline{\lvert\mathcal{M}\rvert^2}_{\mathrm{QCD}}$	$\dfrac{-4g_s^4(\hat{s}^2+\hat{t}^2)}{9\hat{s}\hat{t}}$	Everything combined

Connection to Your Diagrams

Your t-channel diagram (top of page 1): quark p_1 emits gluon p_3 first, absorbs gluon p_2 second.

- Bottom vertex: $c_i \to c_m$ via λ^b
- Top vertex: $c_m \to c_j$ via λ^a
- Color chain: $(\lambda^a\lambda^b)_{ji}$
- Propagator: $1/\hat{t}$

Your s-channel diagram (bottom of page 1): quark p_1 absorbs gluon p_2 first, emits gluon p_3 second.

- Left vertex: $c_i \to c_m$ via λ^a
- Right vertex: $c_m \to c_j$ via λ^b
- Color chain: $(\lambda^b\lambda^a)_{ji}$
- Propagator: $1/\hat{s}$

Different color orderings ($\lambda^a\lambda^b \neq \lambda^b\lambda^a$), but same color factor $32/9$ after summing, because the trace $\mathrm{Tr}[\lambda^a\lambda^b\lambda^b\lambda^a] = \mathrm{Tr}[\lambda^b\lambda^a\lambda^a\lambda^b]$ by the cyclic property.